IPCON® PLA Matte

High Speed Matte Polylactic Acid Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.03 mm
Net Filament Weight	1 kg ± 0.5%
Filament Length	300 ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C)
Spool Size	Outer Cardboard's diameter: 200 mm; Inner roller's diameter: 90 mm

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.39 g/cm³
Shrinkage Rate*	Specially-defined	0.23% - 0.41%
Surface Hardness	Shore hardness	80D
Saturated Water Absorption Rate	25 °C，55% RH，room air	0.43%
Melt Index	210 °C，2.16 kg	17.4 g/10 min
Melting Temperature	DSC，10 °C/min	152 °C
Vicar Softening Temperature	ISO 306，GB/T 1633	60 °C
Heat Deflection Temperature	ISO 75，0.45 MPa	56 °C

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527，GB/T 1040	33 MPa
Tensile Strength Z	ISO 527, GB/T 1040	24 MPa
Young's Modulus XY	ISO 527，GB/T 1040	2874 MPa
Young's Modulus Z	ISO 527，GB/T 1040	2431 MPa
Breaking Elongation Rate XY	ISO 527，GB/T 1040	6.7%
Breaking Elongation Rate Z	ISO 527，GB/T 1040	2.4%
Flexural Strength XY	ISO 178，GB/T 9341	67 MPa
Flexural Strength Z	ISO 178，GB/T 9341	42 MPa
Flexural Modulus XY	ISO 178，GB/T 9341	3330 MPa
Flexural Modulus Z	ISO 178，GB/T 9341	2759 MPa
Impact Strength XY	ISO 179，GB/T 1043	23.8 kJ/m²
Impact Strength Z	ISO 179，GB/T 1043	5.2 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 220 °C, bed temperature: 55 °C, printing speed: XY—270 mm/s and Z—100 mm/s) and IPCON PLA Matte (Can Zi) filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill

pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. Unique 30 kinds of traditional and popular Chinese colors, excellent layer strength and stiffness, able to be printed with “Ludicrous Mode” (compatible with AMS and CFS, max speed is 330 mm/s).
2. Can be used to print collectible toys, statues, model kits, household articles, cultural and creative decors, etc.

• Recommended Printing Settings

Drying Settings Before Printing	55 - 60 °C, 6 - 8 h (Blast drying oven)
Printing Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C，≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself or support filament for PLA
Compatible Printer Type	Open-frame, enclosed-frame (Suggest to open the front door and remove the upper lad)
AMS or CFS	Compatible
Compatible Nozzle Material	Any common material, such as brass, stainless steel, hardened steel, diamond, tungsten carbide, etc
Compatible Nozzle Size	0.2 / 0.4 / 0.6 / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plate
Plate Surface Preparation	Apply liquid glue or PVP solid glue stick if the first layer bonding is not firm enough
Nozzle Temperature	190 - 230 °C
Bed Temperature	35 - 65 °C
Fan Speed	60% - 100%
Max Print Speed	330 mm/s

IPCON Polymer Material (Suzhou) Co., LTD

Address: Room 632, Building E, Dongchuang Science Park, No. 216, Jinfeng Road, Mudu Town,Wuzhong District, Suzhou City,China

Web Site: www.ipconpolymer.com

E-mail: sales@ipconpolymer.com

Tel: 400-139-1152

Innovative pragmatic Consistent